

# Exact exponential rates for the discrete Calderón problem on lattice strips: a computational note

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## Abstract

The preprint cited above found empirically that the condition number of the Jacobian of the Dirichlet-to-Neumann (DtN) map of a resistor network with respect to its edge conductances grows exponentially with the depth of the unknowns on flat lattice disks and polynomially on hyperbolic tilings. Here we determine the exponential rate exactly on a solvable family: a lattice strip, periodic along a boundary row that carries the measurements. Translation invariance makes the Jacobian exactly block-diagonal in the total momentum along the boundary; each block is a Vandermonde-type matrix whose nodes are the products of the per-row decay factors of two boundary modes. The smallest singular value of block  $q$  then decays with depth at the exponent  $G(q)$  of the Green function of the complement of the node set, evaluated on the unit circle, and the global rate is attained at the zigzag momentum  $q = \pi$ . We check this in 140- to 220-digit arithmetic: for the aligned square strip with unit conductances the zigzag rate is 1.653 decades per row, for a boundary along the lattice diagonal it is 1.067, and for lateral-to-vertical conductance ratios 0.05 and 16 it is 1.762 and 2.498 (predicted 1.765 and 2.451, fixed before the computation), non-monotone in the ratio; at converged width all momenta agree with the prediction to within 2 %. The note records its failures: two closed forms of mine were wrong (one preregistered), the diagonal case needed a sign correction that was found after the data, and three design errors (finite reference height, depth windows, tolerances) are kept in a register. The disk, curvature, and complex node sets are open. Nothing here concerns holography.

## 1 Setting

A resistor network has nodes, edges  $e$  with conductances  $g_e > 0$ , and a set of  $m$  boundary nodes where voltages are imposed and currents measured; recovering the conductances from these data is the discrete Calderón problem [1, 2]. The empirical background is the companion preprint [7]. The DtN map  $\Lambda(g) \in \mathbb{R}^{m \times m}$  is the Schur complement of the weighted graph Laplacian onto the boundary. By the envelope theorem the derivative of  $\Lambda$  with respect to the conductance of edge  $e = (a, b)$  is  $\partial\Lambda/\partial g_e = d_e d_e^\top$ , where  $d_e = H_a - H_b \in \mathbb{R}^m$  is the difference of the rows of the harmonic extension  $H$  (boundary data  $\mapsto$  node potentials) at the endpoints of  $e$ . Collecting the strictly upper-triangular entries of  $\partial\Lambda/\partial g_e$  as the column of  $e$  gives the Jacobian  $J$  used throughout (the diagonal of  $\Lambda$  is determined by its row sums). The *depth* of an edge is the smaller of the graph distances of its endpoints to the boundary, and  $J_d$  is  $J$  restricted to the columns of depth  $\leq d$ . We study  $\kappa(J_d) = \sigma_{\max}/\sigma_{\min}$  as a function of  $d$ ; the companion preprint reports  $\log_{10} \kappa(J_d)$  growing roughly linearly with  $d$  on flat disks.

The solvable family is the *strip*: nodes  $(s, j)$ ,  $s = 0, 1, 2, \dots$  (rows,  $s = 0$  the boundary),  $j \in \mathbb{Z}_W$  (periodic along the boundary), semi-infinite in  $s$ . We treat two orientations of the

square lattice relative to the boundary. In the *aligned* strip, node  $(s, j)$  is joined to  $(s, j \pm 1)$  with conductance  $\lambda$  and to  $(s \pm 1, j)$  with conductance 1. In the *diagonal* strip (the lattice rotated by  $45^\circ$  with respect to the boundary, periodic along  $(1, 1)$ ), the rows are  $s = x - y$  and each node of row  $s$  is joined to the nodes  $j$  and  $j + 1$  of row  $s + 1$ , with no edges inside a row; the graph depth is then the  $L^1$  distance to the boundary.

## 2 Block structure

Translation  $j \mapsto j + 1$  acts on the edges and, compatibly, on the data (the strictly upper-triangular entries), and  $J$  intertwines the two actions. Fourier transforming the columns over the position  $j$  of the edge therefore makes  $J$  exactly block-diagonal in the total momentum  $q = 2\pi i/W$ ,  $i \in \mathbb{Z}_W$  (I checked numerically that the union of the block singular values equals the spectrum of the unblocked matrix to  $2 \times 10^{-15}$ ).

**Aligned strip.** A boundary mode  $e^{ikj}$  has the harmonic extension  $z_1(k)^s e^{ikj}$  with  $z_1 = e^{-\kappa(k)}$  and  $\cosh \kappa(k) = 1 + \lambda(1 - \cos k)$ . A vertical edge at depth  $r$  has signature  $d_e \propto z_1(k)^r (1 - z_1(k))$  per mode, so for the vertical-edge columns the block at momentum  $q$  is

$$V_q[n, r] = a_n z_n^r, \quad z_n = z_1(k_n) z_1(q - k_n), \quad a_n = (1 - z_1(k_n))(1 - z_1(q - k_n)), \quad k_n = \frac{2\pi n}{W}, \quad (1)$$

with rows  $n$  indexing the pair of modes  $(k_n, q - k_n)$  and columns  $r = 0, \dots, d$  the depths. Because the data exclude the diagonal of  $\Lambda$ , the position-diagonal direction, which in the block is the uniform vector over  $n$ , is projected out:  $V_q \mapsto (I - uu^\top)V_q$ ,  $u = \mathbf{1}/\sqrt{W}$ . Without this projection the block does not reproduce the explicit Jacobian; with it the exact block agrees with the explicit double-precision block at large height to 0.005 decades in every per-row increment.

**Diagonal strip.** The decay factor  $\lambda(k)$  of the mode  $e^{ikj}$  is the root with  $|\lambda| < 1$  of  $(1 + e^{ik})\lambda^2 - 4\lambda + (1 + e^{-ik}) = 0$ ;  $|\lambda(k)| = \rho(k) = (1 - \sin(|k|/2))/\cos(k/2)$  and the zigzag mode  $k = \pi$  decouples ( $\lambda = 0$ ). The block of the edges joining  $(s, j)$  to  $(s + 1, j)$  has the form (1) with  $z_n = \lambda(k_n)\lambda(q - k_n)$ ,  $a_n = (1 - \lambda(k_n))(1 - \lambda(q - k_n))$ .

## 3 The rate

Let  $K_q \subset \mathbb{C}$  be the set of node values  $\{z_n\}$  of block  $q$ . The geometric growth of the condition number of Vandermonde matrices is classical [5, 6]; what is needed here is its exact rate for the node sets that arise. For a polynomial  $p(z) = \sum_{r \leq d} c_r z^r$  the vector  $V_q c$  has entries  $a_n p(z_n)$ , so  $\sigma_{\min}(V_q^{(d)})$  is governed by how small  $p$  can be on  $K_q$  relative to the coefficient norm  $\|c\|_2$ . The Bernstein–Walsh inequality,  $|p(\zeta)| \leq \|p\|_K e^{d g_K(\zeta)}$  with  $g_K$  the Green function of the complement of  $K$  with pole at infinity [3, 4], gives, with  $|c_r| \leq \max_{|\zeta|=1} |p|$ ,

$$\|c\|_2 \leq \sqrt{d+1} \|p\|_K e^{dG}, \quad G = \max_{|\zeta|=1} g_K(\zeta),$$

hence a lower bound  $\sigma_{\min} \gtrsim e^{-dG}$  up to subexponential factors. For a regular compact set the bound is attained up to subexponential factors by Chebyshev-type polynomials (the standard extremal theory [4]), and the weights  $a_n$  and the one-direction projection only enter the subexponential factors. We use this as the *prediction*

$$\sigma_{\min}(V_q^{(d)}) \approx 10^{-G(q)d}, \quad G(q) = \frac{1}{\ln 10} \max_{|\zeta|=1} g_{K_q}(\zeta), \quad (2)$$

and test it numerically; we do not claim a proof. When  $K_q$  is a real interval  $[a, b]$ ,  $g(\zeta) = \log|w + \sqrt{w^2 - 1}|$  with  $w = (2\zeta - a - b)/(b - a)$  (branch of modulus  $> 1$ ), and in every case

Table 1: Measured / predicted decay rate of  $\sigma_{\min}$  (decades per row) of the momentum blocks, exact semi-infinite computation at  $W = 384$  (last column: zigzag block at  $W = 768$ ). Predictions are the Green-function exponents of the node sets; for the diagonal boundary the signed-node form was found after the data (Section 5), the moduli-only form was the preregistered one.

| Geometry                      | $\pi/6$       | $\pi/3$       | $\pi/2$       | $2\pi/3$      | $5\pi/6$      | $\pi$         | $\pi$ ( $W=768$ ) |
|-------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|-------------------|
| aligned, $\lambda = 1$        | 0.955 / 0.961 | 1.133 / 1.138 | 1.295 / 1.298 | 1.436 / 1.435 | 1.554 / 1.551 | 1.679 / 1.653 | 1.653 / 1.653     |
| diagonal (signed nodes)       | 0.838 / 0.846 | 0.917 / 0.922 | 0.985 / 0.991 | 1.039 / 1.044 | 1.065 / 1.071 | 1.067 / 1.070 | 1.067 / 1.070     |
| aligned, $\lambda = 0.05$     | 1.135 / 1.141 | 1.227 / 1.233 | 1.334 / 1.339 | 1.457 / 1.461 | 1.599 / 1.601 | 1.777 / 1.765 | 1.762 / 1.765     |
| aligned, $\lambda = 16$       | 1.440 / 1.423 | 1.900 / 1.870 | 2.186 / 2.144 | 2.366 / 2.314 | 2.460 / 2.411 | 2.647 / 2.450 | 2.497 / 2.450     |
| diagonal, moduli only (wrong) | 0.838 / 0.849 | 0.917 / 0.938 | 0.985 / 1.036 | 1.039 / 1.144 | 1.065 / 1.265 | 1.067 / 1.403 | 1.067 / 1.403     |

Table 2: Register of the preregistered gates and predictions behind this note, from the stored verdicts. Every refutation is kept; the failed gates are design errors of mine (Section 6).

| Prereg. | Ledger    | Gates                     | Predictions                                       |
|---------|-----------|---------------------------|---------------------------------------------------|
| 23      | H0-X-0014 | G1 pass, G2 pass, G3 fail | P1 refuted, P2 held, P3 refuted, P4 void          |
| 24      | H0-X-0015 | G1 pass, G2 fail          | P1 held, P2 refuted, P3 held                      |
| 25      | H0-X-0016 | G1 pass                   | P1 held, P2 held, P3 held, P4 refuted, P5 held    |
| 26      | H0-X-0017 | G1 pass                   | P1 refuted, P2 refuted, P3 held, P4 held          |
| 27      | H0-X-0018 | G1 fail                   | P1 held, P2 refuted, P3 held, P4 refuted, P5 held |

below the maximum over the circle is attained at  $\zeta = -1$ . Since  $\sigma_{\max}$  is attained at the shallow columns and varies only polynomially with  $d$ , the condition number grows at the largest  $G(q)$ , which is attained at the zigzag momentum  $q = \pi$ : the block with the smallest top node, not the one with the widest spread of nodes. (My first estimate of the rate used the Chebyshev growth of  $[a, 1]$  after rescaling by  $z_{\max}$  and evaluated it at the wrong point; it predicted 1.833 for the aligned strip and was wrong, Section 6.)

A finite boundary carries finitely many data per momentum: at  $q = \pi$  and  $W$  sites the block has only  $W/4 + 1$  distinct nodes (the values are invariant under  $k \mapsto -k$  and  $k \mapsto \pi - k$ ), so  $J_d$  is rank-deficient beyond a depth set by  $W$ , and at depths comparable to the number of distinct nodes the measured per-row loss drifts upward from  $G$ . We therefore take rates over depths 15–25 at  $W = 384$  (97 distinct nodes at  $q = \pi$ ) and check the zigzag block again at  $W = 768$ .

## 4 Verification

**Method.** The blocks (1) are built in 140 digits (220 for the large-ratio case), and  $\sigma_{\min}$  of the first  $d + 1$  columns is computed as  $1/\sqrt{\lambda_{\max}(G^{-1})}$ ,  $G = V^\dagger V$ , by power iteration (the spectrum of  $G^{-1}$  is geometric, so a few iterations suffice). The validity gate compares the exact zigzag block with the explicit double-precision block of the Jacobian of a  $W = 96$  cylinder of large height, for the per-row increments at  $d \leq 5$ .

**Result.** Table 1 and Figure 1 compare measured and predicted rates for six momenta in four geometries. For the aligned strip with  $\lambda = 1$  the zigzag rate is 1.653 at  $W = 768$  (prediction 1.653) and the six momenta agree to 0.06–1.6 % at  $W = 384$ . For the aligned strip with  $\lambda = 0.05$  and 16, whose predictions (1.765 and 2.451 for the zigzag block, and the five other momenta) were fixed before the computation and which have a different dispersion relation, the converged zigzag rates are 1.762 and 2.498; for  $\lambda = 0.05$  all six momenta are within 0.7 %, for  $\lambda = 16$  the five non-zigzag momenta are within 2.2 % (all on the high side) and the zigzag block at  $W = 384$  overshoots by 8 % because its per-row loss still drifts with depth (Figure 2) when only 97 distinct nodes are available; at  $W = 768$  the overshoot is 1.9 %. The zigzag momentum carries the smallest  $\sigma_{\min}$  at depth 20 in every case. The null hypothesis that the rate does not depend on  $\lambda$  is refuted by 6.6 % and 51 %.

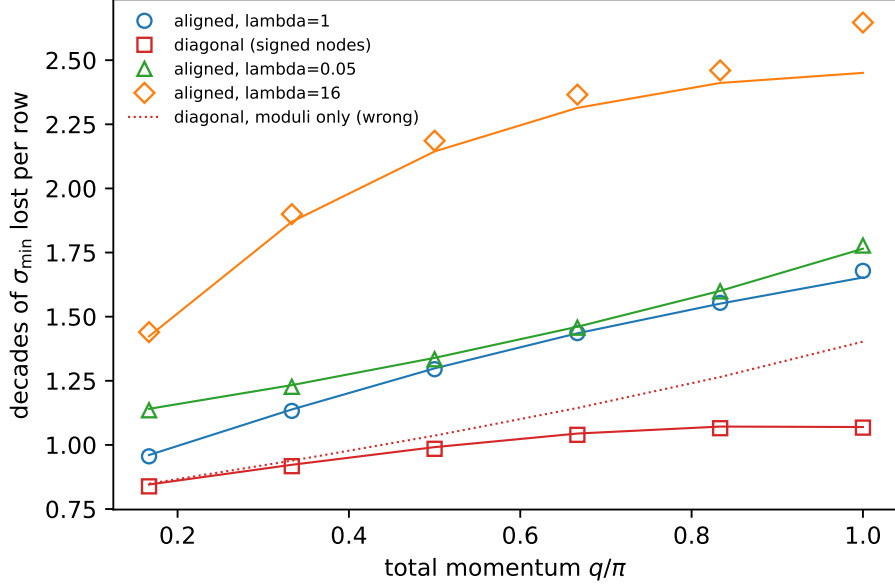


Figure 1: Decay rate of  $\sigma_{\min}$  against total momentum,  $W = 384$ : markers measured, lines predicted (Green exponent of the node set). The dotted red line is the preregistered but wrong prediction for the diagonal boundary (moduli of the nodes only).

## 5 Orientation and signed nodes

For the diagonal strip the exact blocks (gate agreement to 0.001) give the zigzag rate 1.067 decades per row, 35 % below the aligned value, and the same zigzag dominance. Two preregistered forms for this rate were wrong: the Green exponent of the *moduli* of the nodes (1.403) and the geometric guess that the rate per graph step is the aligned rate divided by  $\sqrt{2}$ , because graph depth is the  $L^1$  distance (1.169). The per-row increments printed by the computation alternated with period two. That led to the explanation, *found after the data*: after removing the common phase  $e^{-iq/2}$  the node values  $\lambda(k)\lambda(q-k)$  are real (imaginary parts  $10^{-16}$ ) but of both signs, because the sign flips when  $q-k$  wraps around  $2\pi$ . The node set is then the single interval  $[-M_-, M_+]$  through zero, and its Green exponent reproduces all six measured rates to between 0.2 % and 0.85 % with no free parameter (Table 1). Because this was fitted to the data it is a hypothesis for the diagonal strip, supported by agreement on six numbers; the aligned family of the previous section is the independent test of the underlying picture. For unequal conductances on the two edge types of the diagonal strip the nodes are genuinely complex (imaginary parts up to 0.16 against moduli up to 0.2), and the interval formula does not apply.

## 6 What failed

Every preregistered gate and prediction is kept (Table 2); four of the eleven evaluable predictions of the first three cards were refuted and one was void, and later cards added more. The errors, all mine, in order: (i) the preregistered rate 0.784 for the aligned strip was a single-block estimate and the measurement was 1.78 at finite height; (ii) the corrected closed form 1.833, derived after seeing that number, agreed with a finite-height window average to 3 % and was wrong (the exact value is 1.653); (iii) double-precision data at a finite cylinder height are contaminated by the closed end at the level of tenths of a decade, and the per-row increments were still rising there; a reference gate of height 40 failed at weak lateral coupling, where the explicit increments converge to the exact ones only at height 160; (iv) a depth window of 20–30 at  $W = 96$  is

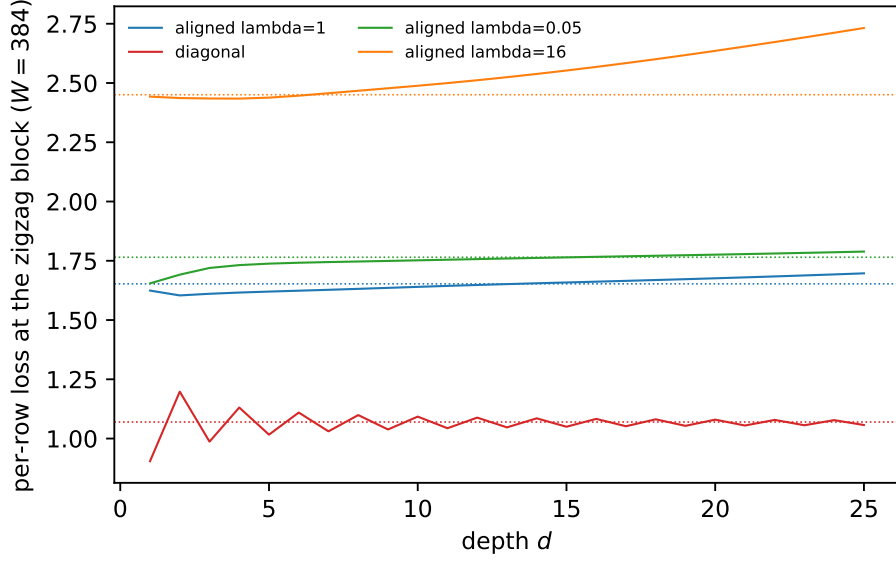


Figure 2: Per-row loss of  $\sigma_{\min}$  at the zigzag block against depth,  $W = 384$  (dotted: prediction). The aligned  $\lambda = 16$  curve drifts upward with depth (too few distinct nodes at this decay rate); the diagonal curve alternates with period two, the signature of nodes of both signs (Section 5).

impossible because the block has 25 distinct nodes; (v) width-convergence tolerances were too tight for fast decay; (vi) the straight periodic triangular boundary row has an exact null vector of the Jacobian at depth 0 (translation invariance plus mirror symmetry), so no triangular value could be obtained on that geometry. These are recorded as lessons in the repository’s lesson file.

## 7 Limits and open questions

All results are computations on the semi-infinite strip with unit or two-valued conductances; the Bernstein–Walsh bound is a sketch, not a proof of the rate. The square disk measured in the companion work loses about 1.1–1.2 decades per graph step, below the aligned strip’s 1.65; a disk combines all boundary orientations with curvature and a staircase boundary, and nothing here says how they combine. The hyperbolic tilings, whose boundary grows with depth, are not covered, nor are triangular lattices with a generic boundary or complex node sets. The finite-height double-precision rates of earlier work should not be read as rates of the semi-infinite problem. The note does not concern holography, AdS/CFT or any physical analogue.

## Reproducibility

Scripts `exp23–exp27` and their preregistrations `PREREGISTRATION_23–27`, the data files under `data/`, the ledger claims H0-X-0014 to H0-X-0018 with evidence digests, and the post-hoc diagnostics (labelled as such) are in the repository of the companion preprint (Zenodo concept DOI 10.5281/zenodo.23000390). Every table and figure of this note is generated from those data files by `paper/make_note_cylinder.py`.

## Use of AI tools

Code, numerical experiments and drafting were carried out with the assistance of Claude (Anthropic); the claims of the later preregistered runs were recorded by a smaller model and audited

against the data files by a larger one, following a written runbook in the repository. The author directed the work and takes responsibility for the content.

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